

WENBIN LIAO

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EDUCATION

University of Michigan

B.S.E. in Computer Science, Minor in Mathematics

Expected Graduation: 2028 (month pending)

Ann Arbor, MI

Relevant Coursework: Probability & Statistics, Linear Algebra, Introduction to Machine Learning, Data Structures and Algorithms, Foundations of Computer Science, Introduction to Computer Organization

SKILLS

Programming: Python, C/C++, SQL, Java, JavaScript/TypeScript

Quantitative & ML: NumPy, pandas, scikit-learn, PyTorch, Statistical Modeling, Cross-Validation, Bootstrap Confidence Intervals, AUROC

Research & Data: Data Cleaning, Feature Engineering, Model Comparison, Experimental Evaluation, Information Retrieval, MapReduce

Systems & Tools: Linux/Unix, Multithreading, Distributed Systems, TCP/UDP, REST APIs, Git/GitHub

EXPERIENCE

Software Developer Intern

IBM Silicon Valley Lab

May 2026 – August 2026

San Jose, CA

- Built **wxO Agent Evaluator**, a Python/FastAPI analytical tool that recursively traverses agent hierarchies, detects configuration failures, and produces structured system-health assessments for complex enterprise AI workflows.
- Engineered a parallel two-pass evaluation pipeline using IBM watsonx.ai, scoring individual nodes and aggregating findings across agent graphs to support repeatable experiments, comparative analysis, and systematic failure diagnosis.
- Designed controlled evaluation workflows to compare agent outputs, isolate failure modes, and validate whether configuration changes improved system behavior rather than introducing regressions.
- Optimized agent discovery using fuzzy matching and TTL-based caching, eliminating repeated API retrievals that could take up to two minutes and improving iteration speed across large environments.
- Automated hundreds of end-to-end QA tests and used structured debugging to reproduce failures, analyze results, identify defects, and validate fixes before release.

Undergraduate AI Researcher

Walbridge Construction | Multidisciplinary Design Program

January 2026 – Present

Ann Arbor, MI

- Built a multi-stage retrieval and reasoning system that maps current project conditions to historical failures, risks, and lessons learned, surfacing relevant evidence from enterprise records for analytical decision-making.
- Ran repeated retrieval experiments and compared relevance, answer accuracy, completeness, and groundedness to distinguish genuine improvements from noisy or inconsistent model behavior.
- Structured orchestration, query, retrieval, filtering, and summarization into independently testable components to isolate failure modes and improve reproducibility across experiments.

Information Technology Intern

Quincy College

July 2023 – August 2023

Quincy, MA

- Performed SQL-based data analysis in the Jenzabar information system using schema-aware queries, data cleaning, and multi-table joins to improve database integrity and reporting accuracy across institutional records.

PROJECTS

Clinical Mortality Prediction | *Python, scikit-learn, NumPy, pandas, Statistical Modeling*

- Built an end-to-end modeling pipeline over noisy ICU records with feature extraction, missing-value imputation, normalization, and stratified 5-fold cross-validation for 30-day mortality prediction.
- Trained and tuned L1/L2 logistic regression and RBF kernel models with class weighting, achieving **0.804 mean CV AUROC** and comparing approaches under a consistent validation framework.
- Evaluated model robustness using AUROC, accuracy, precision, F1, sensitivity, specificity, average precision, ROC curves, bootstrap 95% confidence intervals, and coefficient-based feature analysis.

Electronic Trading Simulator | *C++, Algorithms, Data Structures, OOP/OOD, Linux/Unix*

- Engineered an exchange-style C++ order-matching engine supporting limit, market, timed, and Immediate-or-Cancel orders with **price-time priority** and independent order books across multiple securities.
- Implemented real-time analytics for median trade prices, midpoint pricing, client transfers, commissions, and market statistics during continuous order processing.
- Stress-tested the engine against large order streams, timestamp collisions, malformed inputs, and adversarial edge cases to analyze deterministic correctness, robustness, and performance.

CNN & Vision Transformer Image Classification | *Python, PyTorch, CNNs, Vision Transformers, Transfer Learning*

- Implemented and trained convolutional networks and a Vision Transformer from scratch in PyTorch, including patch embeddings, positional encodings, multi-head self-attention, transformer encoder blocks, and classification heads.
- Designed controlled experiments with cross-entropy loss, Adam, early stopping, and transfer learning; compared model configurations using training/validation loss, accuracy, and AUROC.

Search Engine & Data Retrieval Platform | *Python, MapReduce, TF-IDF, PageRank, Flask, REST APIs*

- Built a scalable retrieval pipeline over a Wikipedia crawl using MapReduce to construct a segmented inverted index and compute TF-IDF weights, document normalization factors, and PageRank scores.
- Implemented distributed Index services that score queries with cosine similarity and merge index-segment scores into final rankings.